International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2022-0146
© 2023 Human Kinetics, Inc.
First Published Online: Jan. 3, 2023

Human Kinetics
ORIGINAL INVESTIGATION

Planning the Microcycle in Elite Football: To Rest or Not to Rest?

Martin Buchheit,[1,2,3] Maxime Settembre,[1] Karim Hader,[1] and Derek McHugh[1]
[1]Kitman Labs, Dublin, Ireland; [2]Lille OSC, Lille, France; [3]HIIT Science, Revelstoke, BC, Canada

Purpose: To examine the association between the programming of days off (ie, no pitch training, days off feet) within turnarounds of varying length and injury rate in elite soccer. ***Methods:*** Retrospective data from 56 team seasons of players belonging to 18 elite teams performing in top leagues including the English Premier League, Italian Serie A, Bundesliga, Scottish Premiership, Major League Soccer, and the Dutch Eredivisie from January 2018 to December 2021 were analyzed (total of 1578 players, 2865 injuries, 2859 noninternational matches, and 12,939 training-session days). The turnarounds examined lasted from 3 to 8 days. Only injuries with ≥3-day time loss were retained for analysis. The authors then looked at the injury rate within each microcycle in relation to the presence of a day off or not and its programming sequences in relation to the previous match (ie, day off at D + 1 vs D + 2 for the day after the match or the following, respectively). ***Results:*** During 3- and 7-day turnarounds, the sequences including the day off feet at D + 2 were associated with 2 to 3 times lower overall noncontact injury rates than the other programming sequences (Cohen *d*: 0.9–2.7). For the other turnarounds, the differences between the sequences were unclear. ***Conclusion:*** The programming of a day off (or at least "off feet") at D + 2 may be associated with moderately to largely lower incidence of noncontact injuries, especially during 3- and 7-day turnarounds.

Keywords: programming, injuries, rest day, off-feet day, soccer

Planning the microcycle is complex in elite football (soccer).[1] While there are some informative data now available on programming practices in soccer,[2–11] these are generally (1) representative of single club practices and only provide quantitative information (eg, external load dynamic based on global positioning system) or (2) limited to periodization guides such as those first developed by R. Verheijen[12] and, later, FC Barcelona.[13] Recently, in order to better understand the reasoning behind the choice and the drivers for planning and content selection, we surveyed 100 elite practitioners working in professional football.[1] The large majority of the responders confirmed collectively that balancing work and recovery from 1 day to the next across the microcycle was very likely required for optimized health and performance.[1] However, whether putting players at complete rest for 1 or 2 days affects injury rate during the same microcycle, and the following match is still unknown. The question of what day to take off, and even whether to give a day off at all is something that has not been examined scientifically despite its immense importance in terms of recovery, compensation, and psychosocial team dynamic.[1,14]

In order to shed light into this important topic, we examined in this descriptive study the association between the programming of days off and injury rate, using retrospective data from 18 elite teams performing in top leagues including the English Premier League (EPL), Italian Serie A, Bundesliga, Scottish Premiership, Major League Soccer (MLS), and the Dutch Eredivisie from January 2018 to December 2021. We more precisely also looked at the timing of these day(s) off within turnarounds of varying lengths. We then looked at the influence of prior match congestions on the abovementioned associations. While the present observational study design precludes the examination of causal relationships, we believe that the information provided could help managers and performance staff to optimize the programming of their microcycles, within their own context.

Methods

Data Collection

For this study, player characteristics, participation data, and injury details were extracted from an online database (ie, Kitman Labs platform Dublin, Ireland) commonly used by all the football teams involved in the study. Each player and club is provided with an ID number on the platform. The researchers in charge of the analysis could only pull and analyze data associated with their IDs—no names included. Data were then transformed and coded for injury occurrence (dates only used for assessing occurrences, such as during a match vs during training and when in relation from/to the previous match) and type (contact or noncontact injury, without any more details), to provide a final data set.

The medical staff of each team registers injury details in the platform as a part of their daily player care management, including variables such as date of injury, type of injury, and injury severity (days lost). Similarly, player game and training session participation are recorded as part of the team staff's daily monitoring. Additionally, the measures of training and competitive load are also added to the platform. The fact that all clubs used the same platform ensured the standardization and the reliability of all types of entries, from medical information to exposure measures (eg, session duration and global positioning system data attached to the system calendar). We, nevertheless, ran a thorough data health check to ensure that all data retained for analysis met the same standard.

Permission was granted by the teams for their inclusion in this study; therefore, ethics committee clearance was not required. The study conforms to the recommendations of the Declaration of Helsinki.

Data were extracted from 18 teams belonging to the English Premier League, Italian Serie A, Bundesliga, Scottish Premiership,

Settembre https://orcid.org/0000-0002-3845-1195
Hader https://orcid.org/0000-0002-2498-6340
McHugh https://orcid.org/0000-0001-5315-030X
Buchheit (mb@martin-buchheit.net) is corresponding author, https://orcid.org/0000-0002-5528-2039

Unauthenticated | Downloaded 01/06/23 01:23 AM UTC

Major League Soccer, and the Dutch Eredivisie from January 2018 to December 2021. This represented 82 team seasons.

Since preliminary analysis did not show any trends suggestive of differences between the different leagues or continents, all data were pooled together to increase sample size.

Team seasons for which injury information was not accessible were not used for analysis. Likewise, when there was not enough information about players on the platform (eg, no exposure for less than 15 players over the entire season), the team season was not included. The final data set included 56 team seasons, including a total of 1578 players, 2865 injuries, 2859 noninternational matches, and 12,939 training session days.

Data Preparation

A n – *d* turnaround was defined as a microcycle with n days between the first and second match, where n is the count of days from the first day after a match up to and including the following match day. The shortest observed turnaround was 3 days, for example, playing a match on Sunday and again the following Wednesday, while the longest was 8 days (8-d turnaround), for example, playing on Saturday and again the following Sunday. In total, 1871 turnarounds were extracted and were grouped by their respective length.

Turnarounds following at least one 3-day turnaround were considered as congested.[15]

In the absence of direct access to teams calendars and schedules, we assumed that a day off was a day without a game where the main 15 players of a given team did not have any training session exposure registered in the Kitman Labs platform. We considered that an exposure took place on a given day when there was information about either workload, duration, or third-party metric with a game or training session event tag. We then considered that these exposures were accurate as they were extrapolated from the metrics consistently collected by the teams. Using this classification, it is very likely that nonexposure days were rest days, but we cannot rule out that some light activities may have taken place at the club (ie, recovery, mobility, football tennis, etc), which, given their nature, were not registered as exposure. Therefore, considering those nonexposure days as "days off-feet" is likely the most accurate description of those specific days—this terminology was consequently used throughout the manuscript.

The 15 outfield players with the highest number of both preseason and in-season games played during a given season were considered as the main players. Note that these 15 main players can be different from one season to another for a given team. Days off-feet distribution patterns were examined within each microcycle. Days were first coded as "x" for a day trained and as "o" for a day off-feet; all possible combinations (eg, x/x/x, o/x/x, x/o/x, x/x/o, o/o/x, o/x/o, x/o/o, o/o/o for 4-d turnarounds) were then created for each turnaround. Only the specific sequences with ≥10 occurrences within each turnaround were retained for analysis.

Injury is often defined as an occurrence sustained during either training or match play which prevents a player from taking part in training or match play for one or more days following the occurrence.[16] In this study, we wanted to focus on noncontact injuries that substantially impact training and match participation and, therefore, only considered noncontact injuries that caused a minimum of 3 days of training/playing interruption, that is, ≥3-day time loss. In fact, we excluded all mild injuries (<2 d lost) because injuries in this category could conceivably not have an impact on the next game availability or training dynamic within the same turnaround. Overall, this choice has allowed us not to include days lost due to potential training removal as a result of player management, as it sometimes happens in clubs.[17] If the medical staff registered injuries from the start to the end of the season, we considered that they strictly did it during the whole season, so we assumed that there was no missing data for this metric in this situation.

Considering all the above, there were 511 main players and 965 time loss injuries, including 559 noncontact ones (both match and training), as part of the 56 team seasons.

Data Analysis

Data were analyzed in 3 consecutive steps, from a macro- to a microlevel.

1. Presence of a day off-feet per se and injury rate: We examined the potential difference in both training and match injury rates (per entire turnaround and per actual training day) with the presence or absence of at least a day off-feet in the turnaround —for all turnarounds pooled together, and then for each specific turnaround length separately.
2. Presence of match congestion (0, 1, or ≥2 of 3-d turnarounds) prior to turnarounds including a least one day off-feet, or not, and injury risk.
3. Distribution (ie, when) of days off-feet during each turnaround length, and their association with training and match injuries.

For the later level of analysis, injuries were presented both per entire turnaround and per actual number of training days only; for example, for x/x/x : overall noncontact injury rate per turnaround was calculated as follows: 0.15 noncontact game injuries per microcycle + 0.05 noncontact training injuries per microcycle = 0.20 noncontact game and training injuries per microcycle; overall noncontact injury rate per training + match days only: 0.15 noncontact game injuries per day + 0.025 (=0.05/2 training days) noncontact training injuries per day = 0.175 noncontact match and training injuries per day.

Statistical Analysis

Results are presented as mean and 95% CIs. Substantial differences were assumed when the CIs did not overlap.[18] Cohen *d* was then calculated to provide a magnitude of the differences, with thresholds of 0.2, 0.6, 1.2, and 2 considered as small, moderate, large, and very large effects/differences.[19]

Results

Overall injury rate was 5 times greater during matches than during training (Table 1), with no difference between turnaround lengths, except for the 5-day turnaround, which displayed fewer injuries than all the others.

Training injuries were slightly lower for the 3- and 4-day turnarounds compared with the longest, but those differences were almost absent when expressed in relation to the actual numbers of days of exposure (Table 1). The differences in both training and match injury rates between turnarounds with and without a day off-feet were unclear. This was observed both when all turnarounds pooled together and also when each specific turnaround was examined separately (all CIs overlapping, data not shown).

The number of congested turnarounds preceding the turnarounds of interest did not have a clear relationship with either training or match injury rate, with or without day(s) off-feet—irrespective of the turnaround length (all CIs overlapping, data not shown).

Unauthenticated | Downloaded 01/06/23 01:23 AM UTC

The most represented training and days off-feet sequences within each turnaround are shown in Table 2. For all turnarounds up to 6 days, the most frequent practice was to train all days of the microcycle (30%–80%, with the shorter the turnaround, the less frequent the days off). For all these turnarounds, if a day off was programmed, it occurred more commonly on D + 1. For the 2 longest turnarounds, 7 and 8 days, the most common practice was to give a day off-feet at D + 1, followed by training every day.

We observed some substantial differences both in noncontact training and match injuries as a function of training and days off-feet sequences within some of the typical turnarounds (Figures 1 and 2). In our sample, there were no noncontact training injuries during 3-day turnarounds when a day off-feet was included (irrespective of the day). The relative frequency of these turnarounds was quite low; however, the match injury rate across the entire 3-day turnarounds with a day off-feet at D + 1

Table 1 Number of Observations for Each Turnaround Length and Associated Overall Training and Match Injury Rate (Irrespective of the Presence of Days Off or Not, Both Contact and Noncontact Injuries Together)

Turnarounds	Turnarounds, n	Training injuries per turnaround	Training injuries/training day	Match injuries
3 d	655	0.05 (0.01)[a]	0.03 (0.01)	0.25 (0.02)
4 d	577	0.06 (0.01)[a]	0.03 (0.01)[b]	0.21 (0.02)
5 d	195	0.13 (0.03)	0.04 (0.01)	0.14 (0.03)[c]
6 d	211	0.22 (0.04)	0.05 (0.01)	0.23 (0.05)
7 d	440	0.18 (0.02)	0.04 (0.01)	0.23 (0.02)
8 d	125	0.23 (0.06)	0.04 (0.01)	0.20 (0.06)

[a]Small substantial difference versus 5- to 8-day turnarounds. [b]Small substantial difference versus 6-day turnarounds. [c]Small substantial difference versus all other turnarounds.

Table 2 Frequency and Proportion of the Most Represented Training and Days-Off-Feet Sequences Within Each Turnaround

Turnaround	Planning sequence	Frequency	Turnaround proportion, %
3 d	o/x/x	94	14
	x/o/x	18	3
	x/x/x	531	80
4 d	o/x/x/x	240	41
	x/o/x/x	28	5
	x/x/x/x	276	47
5 d	o/o/x/x/x	12	6
	o/x/x/x/x	63	29
	x/o/x/x/x	30	15
	x/x/x/x/x	70	34
6 d	o/o/x/x/x/x	25	11
	o/x/x/x/x/x	64	28
	x/o/x/x/x/x	25	11
	x/x/o/x/x/x	11	5
	x/x/x/x/x/x	68	29
7 d	o/o/x/x/x/x/x	53	11
	o/x/x/o/x/x/x	83	17
	o/x/x/x/x/x/x	116	25
	x/o/x/x/x/x/x	44	10
	x/x/x/x/x/x/x	69	15
8 d	o/o/x/x/o/x/x/x	15	12
	o/o/x/x/x/x/x/x	20	15
	o/x/x/x/x/x/x/x	21	16
	x/o/x/x/x/x/x/x	12	9
	x/x/x/x/x/x/x/x	16	12
Totals		2005	

Note: Since some less frequent sequences were not shown here, the proportions (right column) do not always sum up to 100%.

Unauthenticated | Downloaded 01/06/23 01:23 AM UTC

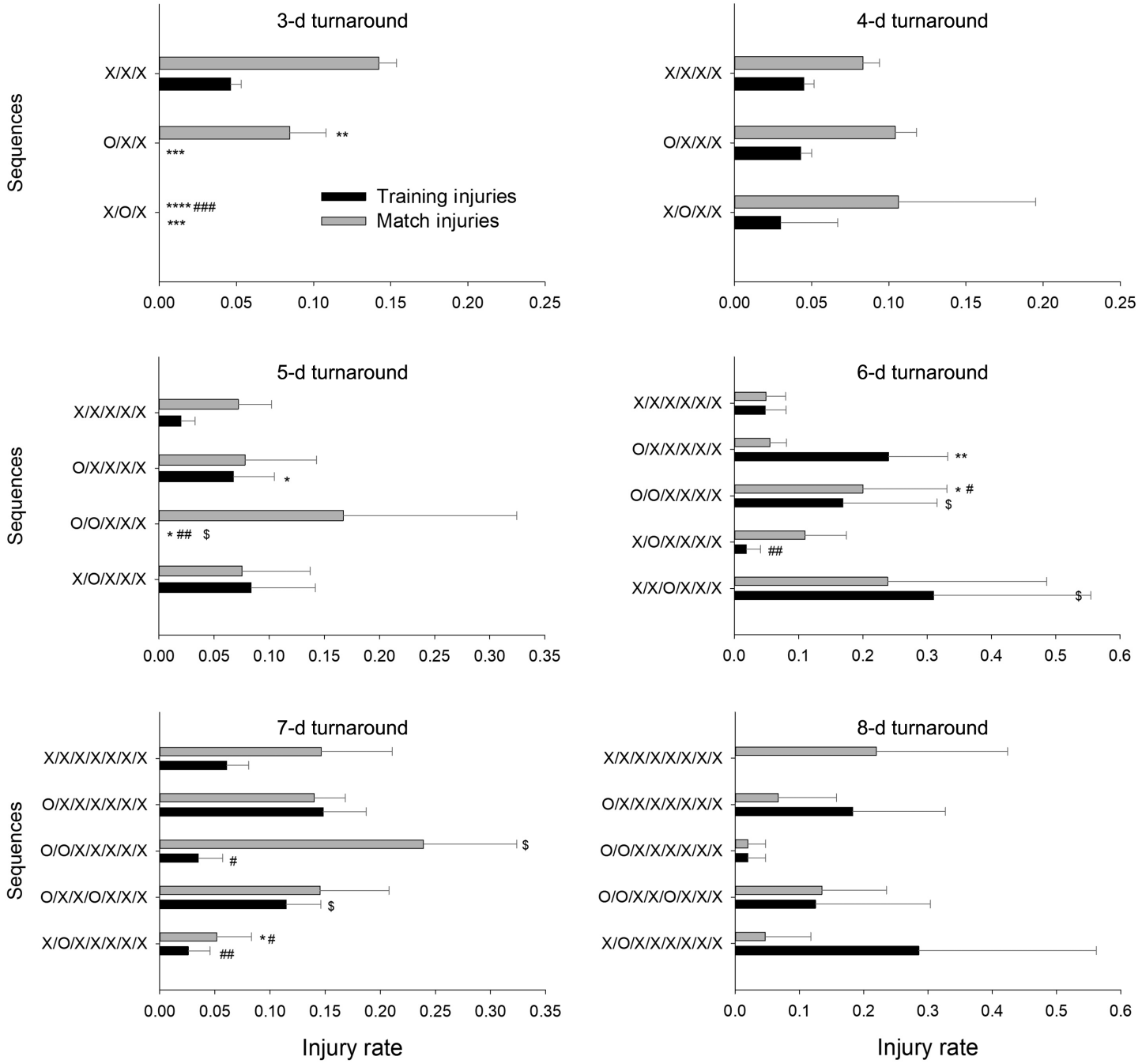

Figure 1 — Average (95% CI) noncontact training and match injury rate during the main training and off-feet-day patterns observed within each of the 6 match turnarounds examined in the 18 teams. *Differences versus x/x/ . . . sequence. #Difference versus o/x/ . . . sequence. $Difference versus x/o/ . . . sequence. The number of symbols stands for small, moderate, large, and very large effects/differences.

was about 50% of the rate for turnarounds with training every day (Table 2). There were no match injuries at all for turnarounds with a day off-feet at D + 2 (Figure 1).

During 5-day turnarounds, there were no noncontact training injuries when 2 days off-feet was programmed at both D + 1 and D + 2. During 6-day turnarounds, the lowest noncontact training injury rate was observed when there was either no day off-feet, and when the latter was programmed at D + 2 (with unclear difference in match injury rate). During 7-day turnarounds, both noncontact training and match injuries were lower when the day off-feet was programmed at D + 2 than when not programmed at all or at D + 1. During 4- and 8-day turnarounds, all injury rate differences between the different sequences were unclear.

When looking at noncontact training and match injuries together (Figure 2), and focusing on the 3 most common programming practices only (ie, no day off, a day off at D + 1 or D + 2), the sequences including the day off at D + 2 (x/o/ . . .) were associated with 2 to 3 times lower injury rates per day (moderate to very large Cohen *d*) than the 2 other sequences for the 3- and 7-day

Unauthenticated | Downloaded 01/06/23 01:23 AM UTC

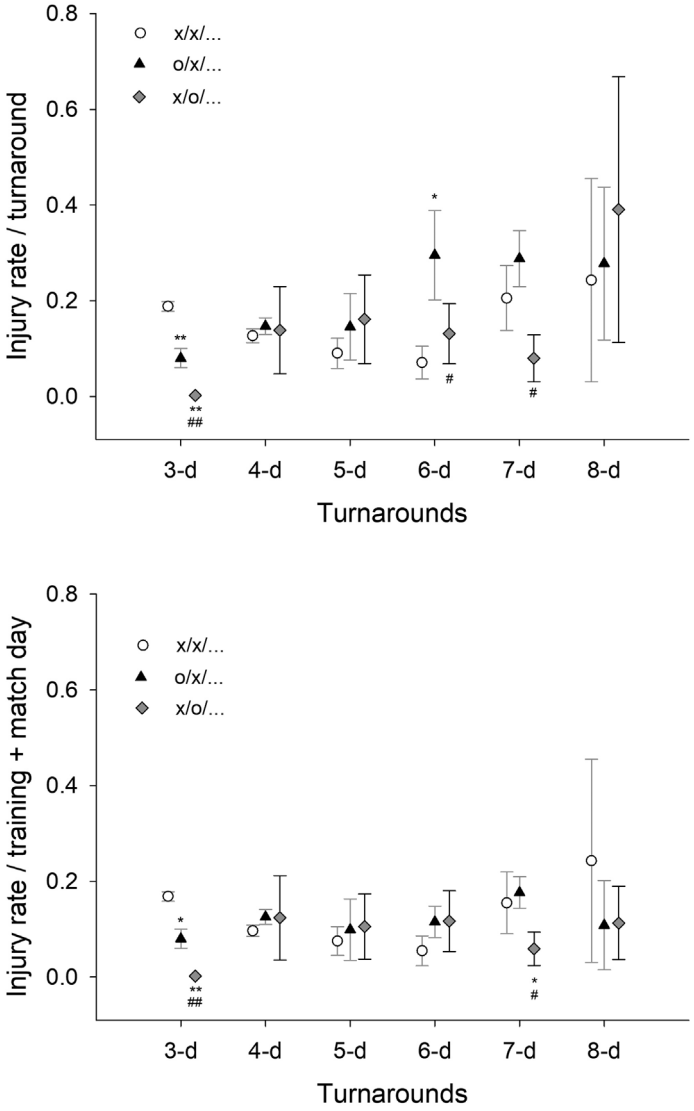

Figure 2 — Average (95% CI) total (training + match) noncontact injury rate per turnaround (upper panel) and total (training + match) noncontact injury rate per training + match days only (lower panel) for the 3 main sequences including either no day off (x/x/ . . .) or a unique day off at either D + 1 (o/x/ . . .) or D + 2 (x/o/ . . .) for all turnarounds. *,**Moderate and large differences versus x/x/ . . . sequence, respectively. #,##Moderate and large differences versus o/x/ . . . sequences, respectively.

turnarounds. For the other turnarounds, the differences between the main 3 sequences were unclear.

Discussion

This is, to our knowledge, the first study to examine the association between the programming of day(s) off (at least days "off feet") within the training microcycle, and both training and match injury rates. While the present observational study design precludes the examination of causal relationships, the present findings suggest that while planning a day off per se may not share clear associations with injury rate (results not shown and Figure 2), the programming and distribution of the day off-feet within the microcycle (ie, when the day off is scheduled), does, especially for 3- and 7-day turnarounds. Despite some variability between the different turnaround lengths, the sequences including the day off-feet at D + 2 (x/o/ . . .) were associated with 2 to 3 times lower injury rates per day (large to very large Cohen *d*) than the 2 other sequences for the 3- and 7-day turnarounds (Figure 2). These associations with injury rate were not affected by prior match congestion, suggestive of the robustness of the association between injury rate and this specific microcycle structure (ie, x/o/ . . .).

While there are always many ways to program the microcycle, training at D + 1 and having a day off at D + 2 may offer several advantages both on the performance and injury sides of things. At D + 1, while the starters of the previous match can receive treatment and perform their recovery session, all benched players and substitutes also have the opportunity to train hard to compensate for the match they did not play. This allows everyone to "close the previous turnaround cycle" (recovery/compensation), and then rest for all the next day (D + 2) before getting back fresh at D + 3 for a new "cycle" until the next match.

Conversely, when having the day off at D + 1, the opportunities to care for starters and compensate for benched and substitute players are reduced, and potentially postponed. The consequence of this is that some starters may still need some treatment at D + 2 and may therefore not be able to train, and subs may have been under a reduced training regime for 2 to 3 consecutive days (light load at D − 1, 0–30 min of play max on match day, and off at D + 1), disturbing an optimal training dynamic and likely limiting their overall adaptation. Along these lines, when training is continually interrupted, substitutes end up lacking training stimulus, especially with respect to some key elements of the game (eg, sprinting distance[20]). They often tend to show reduced neuromuscular performance as the season progresses.[21] While not implying causality, our results may provide support to the common practice of having the (only) day off-feet at D + 2, irrespective of the turnaround length, at least when injury is the consideration.

When looking at specific training and rest days distributions, it appeared that in all turnarounds up to 6 days the most frequent practice was to train on the pitch every day of the microcycle (Table 2). In addition, for all these turnarounds, if a day off-feet was to be programmed, it was preferentially programmed on D + 1. For the 2 longest turnarounds (7 and 8 d; Table 2), the preference was to give a day off-feet at D + 1, followed by training every day. This contrasts with the results of our recent survey of 100 elite practitioners[1] where having the day off at D + 2 (rather than D + 1) was reported to be the optimal option. This may be related to the fact that when responding, the practitioners may have been biased toward their preferences rather than their actual practices (as per the data analyzed in the present study). Therefore, the microcycle structure associated with the lowest injury rate was not the most commonly programmed, irrespective of the turnaround length (eg, the occurrence of the "x/o/x/x/x/x/x" sequence was only 10% vs 25% for the "o/x/x/x/x/x/x" sequence; Table 2). It is also worth noting that coaches may not always want to consider the "injury" argument as their first consideration when programming their microcycles; other factors including psychosocial team dynamics (players generally prefer D + 1 to be off), the need to provide a greater overall training stimulus to players (very little rest or no days off at all during preseason, returning from breaks) or to prepare tactically for important matches, and various external constraints (eg, travels) may often need to be prioritized. Also, while having the day off at D + 2 still allows for a complete training cycle post day off for the longest turnarounds, for example, 4 days

Unauthenticated | Downloaded 01/06/23 01:23 AM UTC

left to prepare the next match during a 7-day turnaround, this may disrupt optimal match preparation during short turnarounds, for example, 1 day left to prepare for the next match during a 4-day turnaround. In summary, coaches may not see the "rest at D + 2 option" as a relevant alternative in their own context even though it may be ideal on paper from a physiological and biological standpoint.

Finally, the reason for the lack of clear and consistent differences in injury rates between the different sequences within the 4-, 5-, 6-, and 8-day turnarounds (Figure 2) is difficult to explain with the current data limited to exposure information. A simple first explanation is likely related to the lower samples for these turnarounds (Table 2), which directly increases the breath of the CIs, making in turn some of the between-sequences differences unclear. It is also likely that other factors may share more associations with injury rate than the programming of days off per se, and, in turn, could have diluted/confounded the analysis. One important limitation is the univariate nature of the present analysis. While we thought to answer the simple question of the programming of rest days, it is clear that injuries are largely multifactorial in nature[22] and that different loading patterns, match exposures and minutes played within the same sequences may also directly affect injury rates. In fact, the data from the practitioners' survey[1] showed that while the current loading and training contents are pretty homogenous between teams for 7-day turnarounds, there is more variability in programming for 5- and 6-day turnarounds. This may partly explain why the association between days off-feet and injury rate was unclear for those turnarounds. We will certainly continue to investigate this and other topics related to planning the microcyle but given the lack of research in this area we encourage other researchers to think about experimental designs, which would provide more insight on how best to adapt the training schedule to the fixture schedule.

In addition, the simultaneous consideration of player profiles (eg, age, injury history, strength, mobility, or flexibility) and other measures of internal training load and responses to load should also improve the analysis—while making the current outputs less straightforward for practitioners. There is in fact a trade-off between the desire for simple questions to have simple answers (eg, when is it best to rest?) and more sophisticated analytic approaches that may have more precision but require more effort to interpret in order to provide direct applications (ie, results of multivariate analyses can be difficult to translate into simple yes/no answers).

Limitations

The present observational study design precludes the examination of causal relationships. Having a proper distinction between complete rest days, days off-feet, and training days would have been ideal. In the absence of direct access to teams' calendars and schedules, days off were estimated based on exposure data (workload, duration, or third-party metric with a game or training session event tag), and we interpreted those days off as at least days off-feet. Whether this perfectly reflects real practices remains impossible to verify, for example, a gym-based session with no measure of external load logged into the system could have been programmed on a day that was counted as "off." It is also worth mentioning that the number of observations for the x/o/x . . . sequences was consistently lower than that for the other sequences, irrespective of the turnaround lengths (Table 2), and this should be considered when interpreting the results. Finally, the injury records used for analysis are as good as what practitioners may have registered. Relying on injuries based on practitioners' entries is however common practice,[23] and we believe that the value of the information provided, derived from a very large sample size (>1800 turnarounds), partly outweighs those possible limitations. Also, the present data showed a 5 × greater injury rate during match than training (Table 1), which is highly consistent with the >24 versus 4 injuries/1000 hours of exposure generally reported.[23] Future research based on more detailed calendar entries and larger sample size for some of the day sequences would help improve the clarity of the current findings.

Practical Applications

The present study showed for the first time, using a large pool of data from elite football, that while planning a day off (at least off-feet) per se may not share clear associations with injury rate, its programming (ie, when) within the microcycle. In practice, at least for the 3- and 7-day turnarounds examined, programming the (only) day off-feet of the microcycle at D + 2 was associated with 2 to 3 times less overall injury rates than either not having a day off-feet, or programming the latter at D + 1. Future studies should also examine, within each turnaround length, the actual load of each training day in relation to the different day off programming strategies.

Conclusion

Programming a day off (or at least "off feet") at D + 2 was associated with a moderate to large reduction of noncontact injuries, especially during 3- and 7-day turnarounds.

Acknowledgments

The authors thank Stephen Smith and Darcy Norman (Kitman Labs) for their comments on a draft of the current manuscript. The authors are all employed by Kitman Labs, a sports performance company used by sports organizations around the world to collect, centralize, and analyze data from multiple sources and aid decision making (www.kitmanlabs.com). There is no direct gain for the authors or Kitman Labs, financial or otherwise, as a result of the findings of this study, or would there be if they had been different. However, the Kitman Labs Performance Intelligence Research Initiative may influence organizations in their choice of technology partner. The authors are bound by an internal research ethics code based on the European Code of Conduct for Research Integrity (https://allea.org/code-of-conduct/). Buchheit was an associate editor of *IJSPP* at the time of submission.

References

1. Buchheit M, Sandua M, Berndsen J, et al. Loading patterns and programming practices in elite football: insights from 100 elite practitioners. *Sport Perf Sci Res*. 2021;153:V1.
2. Castillo D, Raya-González J, Weston M, Yanci J. Distribution of external load during acquisition training sessions and match play of a professional soccer team. *J Strength Cond Res*. 2019;35(12):3453–3458. doi:10.1519/JSC.0000000000003363
3. Chena M, Morcillo JA, Rodríguez-Hernández ML, Zapardiel JC, Owen A, Lozano D. The effect of weekly training load across a competitive microcycle on contextual variables in professional

Unauthenticated | Downloaded 01/06/23 01:23 AM UTC

soccer. *Int J Environ Res Public Health*. 2021;18(10):5091. doi:10.3390/ijerph18105091

4. Clemente FM, Owen A, Serra-Olivares J, Nikolaidis PT, van der Linden CMI, Mendes B. Characterization of the weekly external load profile of professional soccer teams from Portugal and the Netherlands. *J Hum Kinet*. 2019;66(1):155–164. PubMed ID: 30988849 doi:10.2478/hukin-2018-0054
5. Hannon MP, Coleman NM, Parker LJF, et al. Seasonal training and match load and micro-cycle periodization in Male Premier League academy soccer players. *J Sports Sci*. 2021;39(16):1838–1849. PubMed ID: 33759688 doi:10.1080/02640414.2021.1899610
6. Los Arcos A, Mendez-Villanueva A, Martínez-Santos R. In-season training periodization of professional soccer players. *Biol Sport*. 2017;2(2):149–155. PubMed ID: 28566808 doi:10.5114/biolsport.2017.64588
7. Malone JJ, Di Michele R, Morgans R, Burgess D, Morton JP, Drust B. Seasonal training-load quantification in elite English premier league soccer players. *Int J Sports Physiol Perform*. 2015;10(4):489–497. PubMed ID: 25393111 doi:10.1123/ijspp.2014-0352
8. Mateus N, Gonçalves B, Felipe JL, et al. In-season training responses and perceived wellbeing and recovery status in professional soccer players. *PLoS One*. 2021;16(7):e0254655. PubMed ID: 34260665 doi:10.1371/journal.pone.0254655
9. Martín-García A, Gómez Díaz A, Bradley PS, Morera F, Casamichana D. Quantification of a professional football team's external load using a microcycle structure. *J Strength Cond Res*. 2018;32(12):3511–3518. PubMed ID: 30199452 doi:10.1519/JSC.0000000000002816
10. Nobari H, Barjaste A, Haghighi H, Clemente FM, Carlos-Vivas J, Perez-Gomez J. Quantification of training and match load in elite youth soccer players: a full-season study. *J Sports Med Phys Fitness*. 2021;62(4):448–456. PubMed ID: 33768776 doi:10.23736/S0022-4707.21.12236-4
11. Oliveira R, Brito J, Martins A, et al. In-season training load quantification of one-, two- and three-game week schedules in a top European professional soccer team. *Physiol Behav*. 2019;201:146–156. PubMed ID: 30529511 doi:10.1016/j.physbeh.2018.11.036.
12. Verheijen, R. *The Original Guide to Football Periodisation Part 1*. World Football Academy; 2014.
13. Tarragó JR, Massafret-Marimón ML, Seirul·lo F, Cos F. Entrenamiento en deportes de equipo: el entrenamiento estructurado en el FCB [Training in team sports: structured training in the FCB]. *Apunts. Educacion Fisica y Deportes*. 2019;2019(3):103–114. doi:10.5672/apunts.2014-0983.es.(2019/3).137.08
14. Buchheit M. Houston, we still have a problem. *Int J Sports Physiol Perform*. 2017;12(8):1111–1114. PubMed ID: 28714760 doi:10.1123/ijspp.2017-0422
15. Carling C, McCall A, Le Gall F, Dupont G. The impact of short periods of match congestion on injury risk and patterns in an elite football club. *Br J Sports Med*. 2016;50(12):764–768. PubMed ID: 26682867 doi:10.1136/bjsports-2015-095501
16. Bahr R, Clarsen B, Derman W, et al. International Olympic Committee consensus statement: methods for recording and reporting of epidemiological data on injury and illness in sport 2020 (including STROBE Extension for Sport Injury and Illness Surveillance (STROBE-SIIS)). *Br J Sports Med*. 2020;54(7):372–389. PubMed ID: 32071062 doi:10.1136/bjsports-2019-101969
17. Eirale C. Hamstring injuries are increasing in men's professional football: every cloud has a silver lining? *Br J Sports Med*. 2018;52(23):1489 PubMed ID: 29363496 doi:10.1136/bjsports-2017-098778
18. Cumming, G. *Understanding The New Statistics. Effect Sizes, Confidence Intervals, and Meta-Analysis*. Routledge; 2011: 536.
19. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc*. 2009;41(1):3–12. PubMed ID: 19092709 doi:10.1249/MSS.0b013e31818cb278
20. Anderson L, Orme P, Di Michele R, et al. Quantification of seasonal-long physical load in soccer players with different starting status from the English Premier League: implications for maintaining squad physical fitness. *Int J Sports Physiol Perform*. 2016;11(8):1038–1046. PubMed ID: 26915393 doi:10.1123/ijspp.2015-0672
21. Morgans R, Di Michele R, Drust B. Soccer match play as an important component of the power-training stimulus in premier league players. *Int J Sports Physiol Perform*. 2018;13(5):665–667. PubMed ID: 28422525 doi:10.1123/ijspp.2016-0412
22. Fonseca ST, Souza TR, Verhagen E, et al. Sports injury forecasting and complexity: a synergetic approach. *Sports Med*. 2020;50(10):1757–1770. PubMed ID: 32757162 doi:10.1007/s40279-020-01326-4
23. Ekstrand J., Spreco A., Bengtsson H., Bahr R. Injury rates decreased in men's professional football: an 18-year prospective cohort study of almost 12,000 injuries sustained during 1.8 million hours of play. *Br J Sports Med*. 2021;55(19):1084–1092. PubMed ID: 33547038 doi:10.1136/bjsports-2020-103159

Unauthenticated | Downloaded 01/06/23 01:23 AM UTC